

Program 1

1. Design the following code to display "Hello, World!" on the console.

```
// system
using System;

// class
class Program
{
    // main
    static void Main(string[] args)
    {
        // print
        Console.WriteLine("Hello, World!");
    }
}
```

Output:

```
Hello, World!
```

```
...Program finished with exit code 0  
Press ENTER to exit console.
```

Program 2

2. Write a program that demonstrates the usage of different data types, variables, and constants.

```
// system
using System;

// class
class Program
{
    // main
    static void Main(string[] args)
    {
        // variables
        int employees = 50;
        float accuracy = 98.5f;
        double revenue = 1500000.75;
        char grade = 'A';
        bool isActive = true;
        string companyName = "AI Tech Solutions";

        // constant
        const string founder = "Elon AI";
        // print
        Console.WriteLine("Company: " + companyName);
        Console.WriteLine("Founder: " + founder);
        Console.WriteLine("Employees: " + employees);
        Console.WriteLine("Model Accuracy: " + accuracy);
        Console.WriteLine("Revenue: " + revenue);
        Console.WriteLine("Grade: " + grade);
        Console.WriteLine("Active: " + isActive);
    }
}
```

Output:

```
Company: AI Tech Solutions  
Founder: Elon AI  
Employees: 50  
Model Accuracy: 98.5  
Revenue: 1500000.75  
Grade: A  
Active: True  
  
...Program finished with exit code 0  
Press ENTER to exit console. 
```

Program 3

3. Design a C# program that uses different types of operators:

-> **Arithmetic operators**

-> **Relational operators**

-> **Logical operators**

```
// system
using System;

// class
class Program
{
    // main
    static void Main(string[] args)
    {
        // variables
        int projects = 10;
        int completed = 7;

        // arithmetic
        int remaining = projects - completed;
        int total = projects + completed;

        // relational
        bool isMoreWork = projects > completed;
        bool isEqual = projects == completed;

        // logical
        bool isActive = true;
        bool isSuccessful = (completed > 5) && isActive;
```

```
// print
Console.WriteLine("Remaining Projects: " + remaining);
Console.WriteLine("Total Count: " + total);

Console.WriteLine("More Work Left: " + isMoreWork);
Console.WriteLine("Equal Projects: " + isEqual);

Console.WriteLine("Project Successful: " + isSuccessful);
    }
}
```

Output:

```
Remaining Projects: 3
Total Count: 17
More Work Left: True
Equal Projects: False
Project Successful: True

...Program finished with exit code 0
Press ENTER to exit console. 
```

Program 4

4. Design a program to demonstrate the use of control structures in C# (decision-making, loops):

- Use an if-else statement to check whether a number is positive, negative, or zero.
- Use a for loop to display all even numbers from 1 to 100.

Use a while loop to find the factorial of a number..

```
// system
using System;

// class
class Program
{
    // main
    static void Main(string[] args)
    {
        // number
        int num = 5;

        // if-else
        if (num > 0)
            Console.WriteLine("Number is Positive");
        else if (num < 0)
            Console.WriteLine("Number is Negative");
        else
            Console.WriteLine("Number is Zero");

        // for loop
```



```
Console.WriteLine("Even Numbers from 1 to 100:");
for (int i = 1; i <= 100; i++)
{
    if (i % 2 == 0)
        Console.Write(i + " ");
}

Console.WriteLine();

// factorial
int n = 5;
int fact = 1;
int i2 = 1;

// while loop
while (i2 <= n)
{
    fact = fact * i2;
    i2++;
}

// print
Console.WriteLine("Factorial of " + n + " is: " + fact);
}
}
```

Output:

```
Number is Positive
Even Numbers from 1 to 100:
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44
46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84
86 88 90 92 94 96 98 100
Factorial of 5 is: 120

...Program finished with exit code 0
Press ENTER to exit console. 
```

Program 5

5. Create a class Person with properties like Name and Age, and then derive a class Employee that adds an EmployeeID and a Salary. Implement polymorphism to display information for both classes.

```
// system
using System;

// base class
class Person
{
    // properties
    public string Name;
    public int Age;

    // method
    public virtual void Display()
    {
        Console.WriteLine("Name: " + Name);
        Console.WriteLine("Age: " + Age);
    }
}

// derived class
class Employee : Person
{
    // extra properties
    public int EmployeeID;
    public double Salary;

    // override
    public override void Display()
```

```

    {
        Console.WriteLine("Name: " + Name);
        Console.WriteLine("Age: " + Age);
        Console.WriteLine("Employee ID: " + EmployeeID);
        Console.WriteLine("Salary: " + Salary);
    }
}

```

```

// class
class Program
{
    // main
    static void Main(string[] args)
    {
        // object person
        Person p = new Person();
        p.Name = "AI Researcher";
        p.Age = 30;

        // object employee
        Employee e = new Employee();
        e.Name = "AI Engineer";
        e.Age = 28;
        e.EmployeeID = 101;
        e.Salary = 80000;

        // polymorphism
        Person refObj;

        refObj = p;
        refObj.Display();

        refObj = e;
        refObj.Display();
    }
}

```

}

}

Output:

```
Name: AI Researcher
Age: 30
Name: AI Engineer
Age: 28
Employee ID: 101
Salary: 80000

...Program finished with exit code 0
Press ENTER to exit console. 
```

Program 6

6. Design a program to demonstrate exception handling in C# that handles potential errors like division by zero and file not found.

```
// system
using System;
using System.IO;

// class
class Program
{
    // main
    static void Main(string[] args)
    {
        // division
        try
        {
            int a = 10;
            int b = 0;
            int result = a / b;

            Console.WriteLine("Result: " + result);
        }
        catch (DivideByZeroException)
        {
            Console.WriteLine("Error: Cannot divide by zero");
        }

        // file
```

```
try
{
    string data = File.ReadAllText("ai_data.txt");
    Console.WriteLine(data);
}
catch (FileNotFoundException)
{
    Console.WriteLine("Error: File not found");
}

// end
Console.WriteLine("Program continues...");
}
}
```


Output:

```
Error: Cannot divide by zero
Error: File not found
Program continues...

...Program finished with exit code 0
Press ENTER to exit console.█
```

Program 7

7. **Design a simple C# program that demonstrates how to retrieve the metadata of an assembly using the Assembly class.**

```
// system
using System;
using System.Reflection;

// class
class Program
{
    // main
    static void Main(string[] args)
    {
        // assembly
        Assembly asm = Assembly.GetExecutingAssembly();

        // name
        Console.WriteLine("Assembly Name: " + asm.GetName().Name);

        // version
        Console.WriteLine("Version: " + asm.GetName().Version);

        // location
        Console.WriteLine("Location: " + asm.Location);

        // print
        Console.WriteLine("AI Company Assembly Info Displayed");
    }
}
```

Output:

```
Assembly Name: a
Version: 0.0.0.0
Location: /home/a.out
AI Company Assembly Info Displayed
.
...Program finished with exit code 0
Press ENTER to exit console. 
```


Program 8

8. Design a program using a switch statement to determine the day of the week based on the day number (1-7).

Include a default case to handle invalid inputs.

```
// system
using System;

// class
class Program
{
    // main
    static void Main(string[] args)
    {
        // day number
        int day = 3;

        // switch
        switch (day)
        {
            case 1:
                Console.WriteLine("Monday - AI Planning");
                break;

            case 2:
                Console.WriteLine("Tuesday - Model Training");
                break;

            case 3:
                Console.WriteLine("Wednesday - Data Analysis");
                break;
```

case 4:

```
    Console.WriteLine("Thursday - Testing");  
    break;
```

case 5:

```
    Console.WriteLine("Friday - Deployment");  
    break;
```

case 6:

```
    Console.WriteLine("Saturday - Maintenance");  
    break;
```

case 7:

```
    Console.WriteLine("Sunday - Rest Day");  
    break;
```

default:

```
    Console.WriteLine("Invalid Day Number");  
    break;
```

```
    }
```

```
    }
```

```
}
```

Output

```
Wednesday - Data Analysis  
  
...Program finished with exit code 0  
Press ENTER to exit console. 
```


Program 9

9. WAP in java to show the concept of classes and objects..

```
// class
class AICompany
{
    // variables
    String name;
    int employees;

    // method
    void display()
    {
        System.out.println("Company: " + name);
        System.out.println("Employees: " + employees);
    }
}

// class
public class Main
{
    // main
    public static void main(String[] args)
    {
        // object
        AICompany obj = new AICompany();

        obj.name = "AI Tech Solutions";
        obj.employees = 50;

        // call
        obj.display();
    }
}
```

}

}

Output:

```
Company: AI Tech Solutions  
Employees: 50  
  
...Program finished with exit code 0  
Press ENTER to exit console. 
```

Program 10

10. WAP in java to show the concept of methods (Static, Instance).

```
// class
class AICompany
{
    // instance method
    void showInstance()
    {
        System.out.println("Instance Method: AI Project Running");
    }

    // static method
    static void showStatic()
    {
        System.out.println("Static Method: AI Company Info");
    }
}

// class
public class Main
{
    // main
    public static void main(String[] args)
    {
        // object
        AICompany obj = new AICompany();

        // call instance
        obj.showInstance();

        // call static
        AICompany.showStatic();
    }
}
```

}

}

Output:

```
Instance Method: AI Project Running  
Static Method: AI Company Info  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

Program 11

11. WAP in java to show the concept of Constructor and also use THIS keyword.

```
// class
class AlCompany
{
    // variables
    String name;
    int employees;

    // constructor
    AlCompany(String name, int employees)
    {
        // this
        this.name = name;
        this.employees = employees;
    }

    // method
    void display()
    {
        System.out.println("Company: " + name);
        System.out.println("Employees: " + employees);
    }
}

// class
public class Main
```

```
{  
    // main  
    public static void main(String[] args)  
    {  
        // object  
        AICompany obj = new AICompany("AI Tech Solutions", 50);  
  
        // call  
        obj.display();  
    }  
}
```


Output:

```
Project: AI Chatbot  
Project: AI Vision, Team Size: 5  
Project: AI Analytics, Team: 10, Budget: 50000.0  
  
...Program finished with exit code 0  
Press ENTER to exit console.█
```

Program 12

12. WAP in java to show the concept of Method overloading.

```
// class
class AICompany
{
    // method 1
    void project(String name)
    {
        System.out.println("Project: " + name);
    }

    // method 2
    void project(String name, int teamSize)
    {
        System.out.println("Project: " + name + ", Team Size: " + teamSize);
    }

    // method 3
    void project(String name, int teamSize, double budget)
    {
        System.out.println("Project: " + name + ", Team: " + teamSize + ",
Budget: " + budget);
    }
}

// class
public class Main
{
    // main
    public static void main(String[] args)
    {
        // object
```

```
AICompany obj = new AICompany();

// calls
obj.project("AI Chatbot");
obj.project("AI Vision", 5);
obj.project("AI Analytics", 10, 50000);
}
}
```

Output:

```
Working on AI Model Development  
  
...Program finished with exit code 0  
Press ENTER to exit console. 
```

Program 13

13. WAP in java to show the concept of interface.

```
// interface
interface AIWork
{
    // method
    void task();
}

// class
class AIEmployee implements AIWork
{
    // override
    public void task()
    {
        System.out.println("Working on AI Model Development");
    }
}

// class
public class Main
{
    // main
    public static void main(String[] args)
    {
        // object
        AIEmployee obj = new AIEmployee();

        // call
        obj.task();
    }
}
```


Output:

```
AI Company Package Created  
  
...Program finished with exit code 0  
Press ENTER to exit console.█
```

Program 14

14. WAP in java to show the concept of Package Creation.

```
// package
package ai.company;

// class
public class AICompany
{
    // method
    public void display()
    {
        System.out.println("AI Company Package Created");
    }
}

// import
import ai.company.AICompany;

// class
public class Main
{
    // main
    public static void main(String[] args)
    {
        // object
        AICompany obj = new AICompany();

        // call
        obj.display();
    }
}
```


}
}

Output:

```
Error: Cannot divide by zero
Error: File not found
Program continues...

...Program finished with exit code 0
Press ENTER to exit console. 
```

Program 15

15. WAP in java to show the concept of Exception handling.

```
// import
import java.io.*;

// class
public class Main
{
    // main
    public static void main(String[] args)
    {
        // division
        try
        {
            int a = 10;
            int b = 0;
            int result = a / b;

            System.out.println("Result: " + result);
        }
        catch (ArithmeticException e)
        {
            System.out.println("Error: Cannot divide by zero");
        }

        // file
        try
        {
            FileReader file = new FileReader("ai_data.txt");
            System.out.println("File opened successfully");
        }
    }
}
```

```
    }  
    catch (FileNotFoundException e)  
    {  
        System.out.println("Error: File not found");  
    }  
  
    // end  
    System.out.println("Program continues...");  
}  
}
```

Output:

```
Assembly Name: a
Version: 0.0.0.0
Location: /home/a.out
AI Company Assembly Info Displayed

...Program finished with exit code 0
Press ENTER to exit console. 
```